Project Execution Summary

Module 1: Data Collection

- Collected the **Online Retail dataset**.
- Identified important fields such as CustomerID, InvoiceNo, InvoiceDate, Quantity, UnitPrice, and Country.

Module 2: Data Cleaning using Python

- Removed duplicate records.
- Removed records with missing CustomerID.
- Removed cancelled transactions.
- Removed invalid Quantity and UnitPrice values.
- Converted InvoiceDate into the correct date format.
- Calculated **Revenue = Quantity × UnitPrice**.

Module 3: RFM Analysis

- Calculated **Recency, Frequency, and Monetary** values for each customer.
- Created RFM scores for customer analysis.

Module 4: Customer Segmentation

- Classified customers using RFM scores.
- Created four segments:
  - **Champions**
  - **Loyal Customers**
  - **At Risk**
  - **Regular Customers**
- Created the final `customer_segments.csv` file.

Module 5: SQL Server

- Created the **CustomerSegmentationDB** database.
- Created tables for retail transactions and customer segments.
- Loaded the prepared data into SQL Server.
- Created a relationship using **CustomerID**.
- Used SQL for data storage and analysis.

Module 6: Power BI

- Connected Power BI to the **CustomerSegmentationDB** SQL Server database.
- Imported the retail transaction and customer segmentation tables.
- Created a relationship using **CustomerID**.

- Created four KPIs:
    - Total Revenue
    - Total Customers
    - Total Orders
    - Repeat Customer Rate
- Created visualizations for customer segments, revenue, sales trends, and repeat customers.
- Added interactive slicers for **Month, Segment, and Country**.
- Designed an interactive Power BI dashboard with drill-through analysis.

Module 7: Tableau

- Connected Tableau to the prepared retail and customer segmentation data.
- Created a relationship using **CustomerID**.
- Created four KPIs:
    - Total Revenue
    - Total Customers
    - Total Orders
    - Repeat Customer Rate
- Created four visualizations:
    - Revenue by Customer Segment
    - Customer Distribution by Segment
    - Monthly Revenue Trend
    - Repeat Customers by Country
- Added filters for **Segment, Country, and InvoiceDate**.
- Designed the **Customer Segmentation & Retention Analysis Dashboard**.

Module 8: Analysis and Insights

- Compared revenue across customer segments.
- Analyzed customer distribution.
- Studied monthly revenue trends.
- Analyzed repeat customers by country.
- Identified customers requiring retention strategies.

Module 9: Recommendations

- Recommended loyalty programs for Champions and Loyal Customers.
- Suggested targeted offers for At-Risk Customers.
- Suggested promotional strategies for Regular Customers.
- Recommended monitoring repeat customer behavior and revenue trends.

Module 10: Final Project Output

- Completed the **Customer Segmentation & Retention Analysis** project.
- Created dashboards in **Power BI and Tableau**.
- Prepared the final project report containing methodology, analysis, visualizations, insights, recommendations, and conclusion.
- The project demonstrates how data analytics can support **customer retention, revenue growth, and data-driven business decisions**